SALES AUTOMATION + ERP MVP

Product Roadmap & System Sitemap

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1. PROJECT OVERVIEW

Goal

Build a complete:

- Sales Automation Mobile App (Android APK)
- ERP Admin System (Web App)

For:

- FMCG
- Distribution
- Wholesale
- Field Sales Operations

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2. TECHNOLOGY STACK

Layer	Technology
Mobile App	Flutter
ERP Web	React + Vite
Backend	Firebase
Database	Firestore
Authentication	Firebase Auth
File Storage	Firebase Storage
Functions	Cloud Functions
Notifications	Firebase Cloud Messaging
Analytics	BigQuery (Future)
Maps	Google Maps API

Layer	Technology
Face Recognition	ML Kit

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3. SYSTEM ARCHITECTURE

Mobile App

- Offline-first architecture
- Local cache
- Background sync
- GPS tracking
- Real-time updates

ERP Web

- Role-based dashboard
- Analytics
- Reports
- Master management
- Approval workflow

Backend

- Firestore collections
- Cloud Functions
- Event-driven operations
- Notification engine

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4. USER ROLES

Admin

- Full access

General Manager

- Global reports
- KPI monitoring

ASM / Supervisor

- Team monitoring
- Approvals
- Route management

Sales Representative

- Attendance
- Orders
- Collections
- Visits

Distributor

- Order monitoring
- Inventory visibility

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5. MOBILE APP SITEMAP

LOGIN

- Splash Screen
- Login Screen
- Forgot Password
- Device Verification

MAIN NAVIGATION

1. Dashboard

Screens

- KPI Summary
- Monthly Overview
- Timeline Activity
- Visit Summary
- LPC Analytics
- Strike Rate Summary

Widgets

- Total Outlet
- Visited Outlet
- Non Visited Outlet
- Productive Call
- Non Productive Call
- Pending Collection
- Order Summary

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2. Attendance

Screens

- Check In
- Selfie Capture
- Face Verification
- GPS Validation
- Check Out
- Attendance Logs
- Working Hours Summary

Features

- Fake GPS detection
- One attendance per day
- Auto checkout

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3. Route

Screens

- Daily Route
- Route Map
- Route Sequence
- Nearby Outlet
- Route Change Request
- Approved Route

Features

- Geo-fencing
- Route optimization
- Live navigation

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4. Outlet Visit

Screens

- Outlet List
- Outlet Details
- Check In
- Check Out
- Visit History
- Retailer Profile

Features

- GPS validation
- Visit duration
- Outlet classification
- Retailer notes

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5. Order Module

Screens

- Create Order
- SKU Catalogue
- Product Search
- Cart
- Order Summary
- Draft Orders
- Order History

Features

- Barcode scan
- Offline order
- Promotion engine
- Outstanding validation
- Credit limit check
- LPC tracking

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6. Collections

Screens

- Due Invoice List
- Collection Entry

- Payment Mode
- Receipt View
- Collection History

Features

- Cash collection
- Bank collection
- Cheque entry
- Signature capture
- Receipt generation

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7. Reports

Screens

- Daily Sales Report
- Monthly Sales Report
- LPC Report
- Strike Rate Report
- Collection Report
- Attendance Report

Features

- PDF export
- Excel export
- Date filtering

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8. Catalogue

Screens

- Product Categories
- Product Details
- Promotion Offers
- Price List
- New Products

Features

- Product image gallery
- SKU search
- Brand filter

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9. Notifications

Screens

- Alerts
- Approvals
- Announcements
- Reminders

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10. Settings

Screens

- Profile
- Device Info
- Language
- Sync Status
- Logout

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6. ERP WEB SITEMAP

AUTHENTICATION

- Login
- Password Reset
- Session Management

ERP MAIN MENU

1. Dashboard

Sections

- Total Sales
- Active Users
- Live GPS
- Attendance Summary
- Collection Summary
- KPI Charts
- Sales Trend

- Route Efficiency

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2. User Management

Screens

- User List
- Add User
- Edit User
- Assign Role
- Assign Route
- Device Binding
- Face Registration

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3. Outlet Management

Screens

- Outlet List
- Add Outlet
- Geo Tagging
- Outlet Classification
- Outlet Mapping
- Outlet Analytics

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4. Product Management

Screens

- SKU List
- Product Categories
- Brand Management
- Price Management
- Promotion Setup
- Stock Visibility

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5. Route Management

Screens

- Route Planner

- Territory Mapping
- Route Assignment
- Route Optimization
- Route Analytics

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6. Order Management

Screens

- Pending Orders
- Approved Orders
- Invoice Generation
- Order Timeline
- Order Analytics

Features

- Approval workflow
- Discount override
- Order cancellation

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7. Collection Management

Screens

- Outstanding Report
- Collection Entry
- Payment Reconciliation
- Due Aging Report
- Collection Analytics

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8. Attendance Monitoring

Screens

- Live Attendance
- GPS Tracking
- Attendance Logs
- Working Hours
- Attendance Analytics

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9. Inventory Management (Phase 2)

Screens

- Warehouse Stock
- Stock Transfer
- Damage Stock
- Expiry Tracking
- Batch Tracking

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10. Delivery Module (Phase 2)

Screens

- Delivery Queue
- Vehicle Assignment
- Delivery Tracking
- Proof of Delivery

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11. Reports & Analytics

Screens

- Sales Analytics
- LPC Analytics
- Outlet Productivity
- Salesman Performance
- Heatmap Analytics
- Forecasting

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12. Approval Center

Screens

- Route Approval
- Discount Approval
- Credit Approval
- Return Approval
- Expense Approval

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13. Notifications Center

Screens

- Alerts
- Push Notifications
- Broadcast Messages

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14. Settings & Configuration

Screens

- Company Settings
- Tax Settings
- Role Permissions
- Feature Flags
- Backup & Restore

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7. FIRESTORE COLLECTION STRUCTURE

Core Collections

users

- employee profile
- role
- route
- device info

outlets

- outlet profile
- GPS
- route
- retailer info

attendance

- check-in
- check-out
- selfie
- GPS

visits

- outlet visit logs
- timestamps
- GPS

orders

- order header
- totals
- status

order_items

- SKU
- quantity
- pricing

collections

- payments
- receipts

routes

- assigned route
- outlet sequence

products

- SKU master
- pricing
- stock

notifications

- alerts
- approvals

dashboard_stats

- pre-aggregated KPI data

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8. API / EVENT ARCHITECTURE

Core Events

- attendance_submitted
- visit_started
- visit_completed
- order_created
- order_approved
- collection_received
- route_changed
- sync_completed

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9. DEVELOPMENT ROADMAP

PHASE 1 — FOUNDATION

Duration

2 Weeks

Tasks

- Firebase setup
- Firestore schema
- Authentication
- User roles
- App architecture
- ERP boilerplate
- Navigation setup

PHASE 2 — CORE MOBILE MVP

Duration

4 Weeks

Features

- Login
- Dashboard
- Attendance
- Route
- Outlet Visit
- Order Module
- Offline sync
- GPS validation

PHASE 3 — ERP MVP

Duration

3 Weeks

Features

- ERP Dashboard
- User Management
- Outlet Management
- Product Management
- Route Management
- Order Monitoring
- Basic Reports

PHASE 4 — COLLECTIONS & REPORTS

Duration

2 Weeks

Features

- Collections
- Receipt generation
- KPI reports
- PDF export
- Excel export

PHASE 5 — ADVANCED OPERATIONS

Duration

3 Weeks

Features

- Face recognition
- Live GPS tracking
- Notifications
- Approval workflow
- Promotion engine
- Expense module

PHASE 6 — INVENTORY & DELIVERY

Duration

4 Weeks

Features

- Inventory module
- Batch tracking
- Delivery tracking
- Van sales
- Warehouse operations

PHASE 7 — ANALYTICS & AI

Duration

Ongoing

Features

- Forecasting
- AI recommendations
- Heatmaps

- Predictive analytics
- Retailer intelligence

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10. PERFORMANCE STRATEGY

Firestore Optimization

- Small documents
- Pagination
- Composite indexes
- Pre-aggregated dashboard collections
- Offline cache

Mobile Optimization

- Lazy loading
- Image compression
- Background sync
- Battery optimization

ERP Optimization

- Server-side aggregation
- Cached analytics
- Role-based queries

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11. SECURITY STRATEGY

Mobile Security

- Device binding
- Fake GPS detection
- Root detection
- Token validation

Backend Security

- Firestore rules
- Role-based access
- Audit logs
- API validation

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12. MVP PRIORITY

MUST HAVE

- Authentication
- Attendance
- Route
- Outlet Visit
- Order
- Dashboard
- Reports
- Collections

SHOULD HAVE

- Live GPS
- Approvals
- Notifications
- Promotions
- Expense claims

FUTURE FEATURES

- AI forecasting
- Distributor portal
- Retailer app
- Voice order
- Marketplace
- Smart recommendations

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13. FINAL RECOMMENDED ARCHITECTURE

Mobile App

Flutter + Firebase SDK

ERP

React + Vite

Backend

Firebase + Cloud Functions

Database

Firestore

Future Scaling

- PostgreSQL
- BigQuery
- Elasticsearch

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14. SUCCESS FACTORS

The success of this platform depends on:

1. Fast offline sync
2. Simple UX
3. Stable GPS validation
4. Reliable attendance system
5. Scalable Firestore structure
6. Smart aggregation strategy
7. Clean workflow architecture
8. Good reporting
9. Fast loading performance
10. Strong security model

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15. MULTI-COMPANY SAAS ARCHITECTURE

Goal

Sell the platform to multiple companies using a single centralized platform.

This means:

- each company has isolated data
- separate branding
- separate users
- separate subscriptions
- separate permissions

This architecture is called:

Multi-Tenant SaaS

SAAS MODEL TYPES

OPTION 1 — Single Database Multi-Tenant (Recommended MVP)

All companies share:

- same Firebase project
- same Firestore database
- same backend

But every document contains:

```
{
  "tenantId": "company_001"
}
```

Example:

```
orders
  tenantId = companyA

users
  tenantId = companyA
```

Advantages

- cheapest
- fastest
- easiest to maintain

- scalable for MVP

Recommended For

- first 50–100 companies

OPTION 2 — Separate Database Per Company

Each company gets:

- separate Firebase project
- separate database

Advantages

- stronger isolation
- enterprise clients prefer this

Disadvantages

- harder maintenance
- higher cost
- more DevOps work

Recommended For

- enterprise clients
- very large customers

RECOMMENDED SAAS ARCHITECTURE

PHASE 1

Use:

- single Firebase project
- tenantId-based isolation

PHASE 2

Add:

- enterprise isolated deployments

REQUIRED MULTI-TENANT DATABASE DESIGN

Every important collection MUST contain:

```
{
  "tenantId": "tenant_001"
}
```

Required Collections

- users
- outlets
- attendance
- visits
- orders
- collections
- routes
- products
- reports
- notifications

FIRESTORE SECURITY RULES

Critical requirement.

Users MUST only access their tenant data.

Example logic:

```
request.auth.token.tenantId == resource.data.tenantId
```

This is extremely important.

SAAS SUPER ADMIN PANEL

You need an additional panel for yourself.

Super Admin Features

Company Management

- create company
- suspend company
- activate company
- delete company

Subscription Management

- package control
- payment tracking
- renewal reminders
- feature access

Usage Monitoring

- total users
- total orders
- storage usage
- API usage

White Label Control

- company logo
- company colors
- app name
- domain mapping

COMPANY ONBOARDING FLOW

Step 1

Create company

Step 2

Generate tenantId

Step 3

Create admin account

Step 4

Assign subscription

Step 5

Initialize default settings

SAAS SUBSCRIPTION PLANS

BASIC

- small team
- limited users
- basic reports

PROFESSIONAL

- advanced analytics
- collections
- GPS tracking
- approvals

ENTERPRISE

- AI features
- dedicated database
- advanced integrations
- custom branding

FEATURE FLAG SYSTEM

Enable/disable modules per company.

Example:

```
companyA
  collections = true
  inventory = false

companyB
  collections = true
  inventory = true
```

Very important for SaaS.

WHITE LABEL SUPPORT

Allow:

- company logo
- splash screen
- app colors
- custom APK
- company name

This increases commercial value greatly.

CUSTOM DOMAIN SUPPORT

Example:

```
companyA.yourplatform.com
companyB.yourplatform.com
```

Future enterprise feature.

BILLING & PAYMENT MODULE

Track:

- subscription status
- invoices

- payment history
- renewals
- trial periods

MULTI-TENANT PERFORMANCE STRATEGY

Important Rules

NEVER query without tenantId

BAD:

```
orders collection full scan
```

GOOD:

```
WHERE tenantId == currentTenant
```

INDEX STRATEGY

Create composite indexes using:

- tenantId
- createdAt
- userId
- routeId

This keeps SaaS performance fast.

MULTI-TENANT ANALYTICS

Track per company:

- sales volume
- active users
- sync health
- attendance quality
- storage usage

FUTURE SAAS SCALING

Future Architecture

Purpose	Technology
Mobile Live Data	Firestore
ERP Analytics	PostgreSQL
BI & Forecasting	BigQuery
Search	Elasticsearch
Media Storage	Firebase Storage

RECOMMENDED COMMERCIAL MODEL

One-Time License

OR

Monthly Subscription (Recommended)

Example:

- per user pricing
- per company pricing
- feature-based pricing

SaaS subscription is much more scalable.

MOST IMPORTANT SAAS REQUIREMENTS

1. Strong tenant isolation
2. Feature flag system
3. Scalable Firestore indexes
4. Billing system
5. White-label support
6. Super admin panel
7. Audit logs

8. Usage tracking
9. Fast onboarding
10. Secure Firestore rules

FINAL SAAS RECOMMENDATION

For your platform:

BEST STARTING ARCHITECTURE

Frontend

- Flutter Mobile App
- React ERP Web

Backend

- Firebase + Cloud Functions

Database

- Firestore

SaaS Model

- Single database multi-tenant
- tenantId isolation

Scaling Later

- PostgreSQL
- BigQuery
- Elasticsearch

This is the best balance between:

- speed
- cost
- scalability
- maintainability
- MVP launch speed